

Proof audit report: `hub_removal.tex`

proof-checker run 01, 2026-09-28

WARNING

Reviewer backend unavailable. The skill requires an adversarial review by an external model through the Codex MCP server. That server is not connected in this session. Review rounds 1 and 2 were therefore self-reviews by the executing agent, using the mandatory checklist A–H and the two-axis severity system.

- Following the skill’s verdict table, the machine-readable verdict is **ERROR** (`reviewer_error`).
- The self-review acceptance gate is **PASS**.
- A cross-model verdict requires re-running the skill with a Codex reviewer connected.

Input. Commit 055505f, sha256 4d562ccc...0090.

Output. The fixed file has sha256 bc8fd63c...62df; it has 17 pages and compiles cleanly.

Companion files. `PROOF_SKELETON.md` (ledger), `PROOF_AUDIT.md` (round log), `PROOF_AUDIT.json`, and [.aris/traces/proof-checker/2026-09-28_run01/](#) (scripts and outputs).

1. Overview

ID	Severity	Status / Impact	Category	Fix strategy	Result
R1-01	MAJOR	UNJUSTIFIED / LOCAL	SCOPE.OVERCLAIM	WEAKEN_CLAIM	fixed (F1)
R1-02	MAJOR	UNJUSTIFIED / LOCAL	SCOPE.OVERCLAIM	WEAKEN_CLAIM	fixed (F2)
R1-03	MAJOR	UNJUSTIFIED / LOCAL	REFERENCE.MISMATCH	ADD_REFERENCE	fixed (F3)
R1-04	MAJOR	UNJUSTIFIED / LOCAL	QUANTIFIER.ERROR	ADD_DERIVATION	fixed (F4)
R1-05	MINOR	UNCLEAR / COSMETIC	CASE.INCOMPLETE	ADD_DERIVATION	fixed (F5)
R1-06	MINOR	UNCLEAR / COSMETIC	forward reference (acyclic)	ADD_DERIVATION	fixed (F6)
R1-07	MINOR	UNCLEAR / COSMETIC	symbol reuse	notation	fixed (F7)
R1-08	MINOR	OVERSTATED / COSMETIC	SCOPE.OVERCLAIM	WEAKEN_CLAIM	fixed (F8)
R1-09	MINOR	UNCLEAR / COSMETIC	CONSTANT_	notation	fixed (F9)
R1-10	MINOR	UNCLEAR / COSMETIC	DEPENDENCE.HIDDEN	WEAKEN_CLAIM	fixed (F10)
R1-11	MINOR	UNCLEAR / COSMETIC	UNJUSTIFIED_	WEAKEN_CLAIM	fixed (F10)
R1-12	MINOR	UNCLEAR / COSMETIC	ASSERTION (in a sketch)	WEAKEN_CLAIM	fixed (F10)
R1-13	MINOR	UNCLEAR / COSMETIC	NORMALIZATION_	notation	fixed (F11)
R2-01	MINOR	UNCLEAR / COSMETIC	MISMATCH	notation	fixed (F11)
			STOCHASTIC_MODE_	notation	fixed (F12)
			CONFUSION	notation	fixed (F12)
			CONSTANT_	notation	fixed (F13)
			DEPENDENCE.HIDDEN	notation	fixed (F13)
			CASE.INCOMPLETE (in F10’s wording)	WEAKEN_CLAIM	fixed in round 2

Totals: 0 FATAL, 0 CRITICAL, 4 MAJOR, 10 MINOR; all 14 were fixed and none remains open. No issue affects the proof of Theorem 1.2, Theorem 1.3, or Corollaries 1.4–1.6.

2. Logic chain of the main result

KEY INSIGHT

The main dependency chain is unchanged by the audit.

- *Upper bound:* Lemma 3.1 (fibres) $\Rightarrow$ Proposition 3.2, which gives $\Phi_K \leq \Psi(N, B)$.
- *Lower bound, first half:* Lemma 4.1 (robust edges) together with Lemma 4.2 (doubling) gives Corollary 4.3, the SCC $\Sigma \supset S_M$.
- *Lower bound, second half:* Lemma 4.4 (attachment) together with Lemma 4.5 (arithmetic core) gives Proposition 4.6, whose error is non-asymptotic.
- *Assembly:* these give Theorem 1.3, hence Theorem 1.2; then Corollary 1.4 follows via Lemma 2.7, which is the only new node.

Every edge of this chain was re-derived. Every application discharges its hypotheses (ledger, `PROOF_SKELETON.md` §2). The dependency graph is acyclic.

3. The fixes

F1 (R1-01): the explicit form for $K \leq \sqrt{N} - 1$

BEFORE

“So the expected fraction of vertices lost ... is asymptotically $\log \frac{\log N}{\log(N/K)}$, which is about $\frac{\log K}{\log N}$ when $K = N^{o(1)}$.”

WHY WRONG

Remark 5.1 proves only $|1 - \mathbb{E}\Phi_K/N - \log \frac{\log N}{\log B}| \leq 31/\log \log N$. The main term is of order $\log K/\log N$. So a ratio asymptotic follows only when $\log K \gg \log N/\log \log N$, and not for bounded K .
Candidate test at $K = 1$: the main term is $\approx 0.693/\log N$ and the proven lower bound on the loss is $\approx 0.667/\log N$. Neither confirms nor refutes the ratio claim.

AFTER

“Up to an additive error $O_\rho(1/\log \log N)$ that is uniform in $0 \leq K \leq \sqrt{N} - 1$, the fraction lost equals $\log \frac{\log N}{\log B}$. This main term dominates the error only when $\log K \gg \log N/\log \log N$; for bounded K we do not determine the loss to within a factor $1 + o(1)$.”

F2 (R1-02): degree attacks at $\theta = 1$

BEFORE

The abstract states the profile for $0 \leq \theta \leq 1$, then: “The same limit holds if the K vertices of largest degree are deleted instead.”

WHY WRONG

Corollary 1.6 assumes $\theta < 1$. Its sandwich lemma needs $N/K \gg \tau_N$, which fails for $K = N^{1-o(1)}$. The claim at $\theta = 1$ is unproven, though plausible for static attacks.

AFTER

“For $0 \leq \theta < 1$ the same limit holds if the K vertices of largest degree are deleted instead.”

F3 (R1-03): prime-counting asymptotics

BEFORE

$\pi(N/2) - \pi(N/3) \sim N/(6 \log N)$ and $\pi(N) - \pi(N/2) \sim N/(2 \log N)$, with only Chebyshev's upper bound cited.

AFTER

Both asymptotics are attributed to the prime number theorem (Hadamard 1896; de la Vallée Poussin 1896), and both works are added to the bibliography. The accompanying inequalities $\geq \pi(N/2) - \pi(N/3)$ and $\geq \pi(N) - \pi(N/2)$ are exact and unchanged.

F4 (R1-04): uniformity in $\rho \in [\rho_0, 1 - \rho_0]$

BEFORE

"The proof depends on ρ only through q and λ . So Theorems 1.2 and 1.3 hold uniformly for $\rho \in [\rho_0, 1 - \rho_0]$, with $N_0 = N_0(\rho_0)$."

WHY WRONG

In the proof of Theorem 1.3, s_0 is defined from $\lambda(\rho)$. The text never explains why a single N_0 works for the whole interval.

AFTER

Put $\lambda_0 := 1 - \rho_0 \geq \lambda$ and $q_0 := \rho_0(1 - \rho_0) \leq q$, and define s_0 from λ_0 .

- Then $\lambda^{s_0} \leq \lambda_0^{s_0} \leq 1/\ell$.
- Conditions (R1)–(R4) and (R6) depend only on N and $s_0(\lambda_0)$.
- The left side of (R5) is decreasing in q once $N^{2\delta} \geq 2$, so (R5) for q_0 implies it for every $q \geq q_0$.

Hence one $N_0(\rho_0)$ works for the whole interval.

F5 (R1-05): the case $t < 0$ in Lemma 4.1

BEFORE

"The second inequality uses $M \leq N$ and $1 - q \leq e^{-q}$."

WHY WRONG

The inequality $(1 - q)^t \leq e^{-qt}$ requires $t \geq 0$, and it reverses for $t < 0$, that is, when $N^{2\delta} < 2$. The *stated* bound still holds in that case, because its right-hand side exceeds 1.

AFTER

"... and $(1 - q)^t \leq e^{-qt}$ for $t = N^{2\delta} - 2 \geq 0$. If $N^{2\delta} < 2$, the right-hand side exceeds 1 and there is nothing to prove."

F6 (R1-06): forward reference replaced by Lemma 2.7

BEFORE

Corollary 3.3 cited "the computation of $\lim \Psi(N, B)/N$ in the proof of Corollary 1.4". That proof opens by invoking Theorem 1.2.

AFTER

Lemma 2.7 (new). If $\log(K + 1)/\log N \rightarrow \theta \in [0, 1]$, then $\Psi(N, N/(K + 1))/N \rightarrow \rho_{\text{Dick}}(1/(1 - \theta))$.

The proof is the former computation, with one added clause: for $\theta < 1$ we have $y > 1$ for large N , so u_N is defined. The lemma uses only Lemmas 2.2 and 2.3. Corollary 3.3 now follows from Proposition 3.2 and Lemma 2.7, and Corollary 1.4 from Theorem 1.2 and Lemma 2.7.

F7–F13 and R2-01: exposition

AFTER

- **F7:** $\epsilon \rightarrow \kappa \in (0, \frac{1}{2})$ in Corollary 1.6, with $1 \leq K$ made explicit and the note $N/(K(K + 1)) \geq \frac{1}{2}N^{2\kappa}$ added.
- **F8:** “far more damaging” is restricted to moderate θ , with the example $\theta = \frac{1}{2}$: half the edges go, but only 30.7% of the vertices remain in the giant.
- **F9:** Remark 5.2 is restated as $\liminf \mathbb{E}\Phi_K^{(2)}/K \geq \frac{1}{2}$, with the dependence of its $o(1)$ terms declared. Theorem 1.2 with $K = 0$ is noted as an internal replacement for [KP, Cor. 2].
- **F10 and R2-01:** the sketch in Remark 7.2 now says “if the p_i are distinct, $s(m) \geq 3$ ”.
- **F11:** “ $K = N^{\theta+o(1)}$ ” is glossed as $\log(K + 1)/\log N \rightarrow \theta$ in the abstract and in the caption of Table 1.
- **F12:** “w.h.p. up to $o(N)$ ” is quantified: for every fixed $\epsilon > 0$, with probability $1 - o(1)$, by Markov’s inequality.
- **F13:** Theorem 1.3 is non-trivial only when $\log \log N > 30$; this is now stated.

4. Proof-obligation diff

Obligation	Before	After
Uniform N_0 for $\rho \in [\rho_0, 1 - \rho_0]$	asserted	derived (F4)
PNT-based asymptotics in §1.4 and (O1)	uncited	cited (F3)
Ratio asymptotic for the fraction lost	claimed	withdrawn; the additive form is proved (F1)
Degree attacks at $\theta = 1$	claimed in the abstract	withdrawn; now open (F2)
$\lim \Psi(N, N/(K + 1))/N$	inside a later proof (forward reference)	standalone Lemma 2.7 (F6)
Lemma 4.1 for $N^{2\delta} < 2$	implicit	explicit (F5)
Every other ledger entry	discharged	discharged (unchanged)

5. Counterexample red team

No counterexample was found. The suite, whose scripts are in the trace directory, was run before the fixes and again after them, with identical results.

• Exhaustive checks:

- Lemma 3.1 and Proposition 3.2 for $N \leq 3000$ and the edge values of K ;
- Lemma 4.2 on 1,643 cases, plus 20 exact boundary cases with $M = 4(K + 1)$;
- Lemma 4.5(c) on 67,921 integers, including the extremes $K = \lfloor N^{1-\eta} \rfloor$ and $\delta = \eta/8$;
- the identity in Remark 5.1;
- Lemma A.1: 20,000 random cases and all 280 grid partitions at $\theta = \frac{1}{3}$, in exact rational arithmetic.

- **Sharpness:**
 - the bound $|V_n| \leq K$ in Lemma 3.1(c) is attained;
 - $\min s(m) = s_0 + 1$ in Lemma 4.5(c) is attained;
 - beyond its hypothesis, the static sandwich of Lemma 6.2 first fails at $K = 833$ ($N = 10^5$), well past the $K \leq 187$ that the hypothesis allows.
- **Extreme parameters:** Lemma 2.4 holds at $z = 2$ and at $z = N$.
- **Constants:** the grid requires $c_0 \geq -0.71$, while the Mertens constant is at least 0.144.
- **Regression trend for Lemma 2.7:** $\Psi(N, N^{1/u})/N - \rho_{\text{Dick}}(u)$ decreases steadily for $N = 10^4, \dots, 10^7$ at $u \in \{1.5, 2, 3\}$.
- **Not falsifiable numerically:** Theorem 1.3 is vacuous below $N \approx 10^{4.6 \cdot 10^{12}}$ (recorded as R1-13).

6. Summary

- **Now proven (unchanged by the audit):**
 - Theorem 1.2 and Theorem 1.3, uniform in K ;
 - Corollary 1.4, the profile $\rho_{\text{Dick}}(1/(1 - \theta))$ for $\theta \in [0, 1]$;
 - Corollary 1.5, the structure of the giant;
 - Corollary 1.6, degree attacks for $\theta < 1$;
 - Remarks 5.1–5.3;
 - Lemma 2.7, newly isolated.
- **Assumed (cited):** Mertens’ theorems, Chebyshev’s bound, Dickman’s asymptotic with Hildebrand’s uniformity, the prime number theorem, Bertrand’s postulate, and [KP, Cor. 2]. The last is replaceable by Theorem 1.2 with $K = 0$.
- **Still a sketch:** Remark 7.2 (hub cliques lose for $\theta > \frac{1}{3}$); it is not used anywhere.
- **Open:**
 - (O1)–(O5);
 - degree attacks at $\theta = 1$;
 - the ratio asymptotic for the fraction lost when K is bounded.